

Two-layer prediction model

Layer 1: pure roofline

AI = FLOPs / bytes; peak roofs

Blind to:

- ✗ L2 cache (hides HBM traffic)
- ✗ per-CTA schedule / occupancy
- ✗ pipeline fill (work-starvation)

Wrong regime at every
single-stream decode step.

Layer 2: per-CTA-corrected

same AI + the schedule

Adds:

- ✓ smem/TMEM → occupancy
- ✓ GEMV vs GEMM (M=1 / 128)
- ✓ pipeline depth; counter-prediction

**Matched the limiter
at all 7 decode steps.**

Both predictions are pre-registered before coding;
the gap between the layers is the finding.